

# End-to-end Keyword Spotting using Neural Architecture Search and Quantization

David Peter, Wolfgang Roth, Franz Pernkopf

Signal Processing and Speech Communication Laboratory  
Graz University of Technology  
Graz, Austria

david.peter@student.tugraz.at, roth@tugraz.at, pernkopf@tugraz.at

## Abstract

This paper introduces neural architecture search (NAS) for the automatic discovery of end-to-end keyword spotting (KWS) models in limited resource environments. We employ a differentiable NAS approach to optimize the structure of convolutional neural networks (CNNs) operating on raw audio waveforms. After a suitable KWS model is found with NAS, we conduct quantization of weights and activations to reduce the memory footprint. We conduct extensive experiments on the Google speech commands dataset. In particular, we compare our end-to-end approach to mel-frequency cepstral coefficient (MFCC) based systems. For quantization, we compare fixed bit-width quantization and trained bit-width quantization. Using NAS only, we were able to obtain a highly efficient model with an accuracy of 95.55% using 75.7k parameters and 13.6M operations. Using trained bit-width quantization, the same model achieves a test accuracy of 93.76% while using on average only 2.91 bits per activation and 2.51 bits per weight.

**Index Terms:** keyword spotting, neural architecture search, quantization

## 1. Introduction

Automatic speech recognition (ASR) is becoming increasingly important for user interaction with everyday consumer devices. ASR systems are typically complex and computation-intensive, i.e. running ASR in always-on mode results in a steady high energy consumption. This is especially problematic for mobile devices whose batteries are drained quickly when running ASR permanently.

A common solution is to run a low-cost keyword spotting (KWS) system that is listening permanently only for a limited set of prespecified keywords. Upon detection of such a keyword, a full ASR system is triggered which then listens for a rich set of user commands. The requirements of a KWS system are: (i) The system should be resource-efficient to mitigate the aforementioned energy problem, (ii) it should run in real-time and, (iii) it should be accurate to maintain a high user-experience.

In KWS, deep neural networks (DNNs) have become the state-of-the-art. In [1] several models from the literature [2, 3, 4, 5] are evaluated on the Google speech commands dataset [6]. They compare their models in terms of accuracy, memory requirements and number of operations (i.e. the number of multiplications and additions) per forward pass. To allow for easy deployment on microcontrollers they train their models using 32 bit float numbers and quantize the weights after training to 8 bit fixed-point numbers. They argue that fixed point numbers have been shown to suffice to run DNNs with minimal

loss in accuracy [7, 8, 9].

There are many aspects to consider when designing resource-efficient DNNs for KWS. In this paper, we will focus specifically on the following important aspects: The DNN architecture, comparison between spectral MFCC feature and raw audio processing as well as quantization of weights and activations.

Recently, neural architecture search (NAS) became a popular technique to automate the design of DNN architectures. A NAS algorithm searches for the best performing DNN architecture within a given search space using appropriate search heuristics. Popular NAS approaches use concepts such as reinforcement learning [10], gradient based methods [11] or evolutionary methods [12] for exploring the search space. In the context of resource-efficient DNNs, NAS techniques have also been used to find DNN architectures that are specifically tailored to the underlying hardware [13, 14] by, for instance, additionally minimizing memory requirements, number of operations, or latency of the resulting model. Therefore, NAS techniques are well-suited for finding DNNs that run on mobile phones or embedded devices. In the context of KWS, NAS has been used successfully by [15, 16] to find efficient and small DNN architectures.

KWS with DNNs is typically performed on hand-crafted speech features such as MFCCs that are extracted from raw audio waveforms. Extracting MFCCs involves computing the Fourier transform. However, performing the Fourier transform is computationally expensive and might exceed the capabilities of resource-constrained devices. Therefore, Ibrahim et al. [17] proposed to use simpler speech features derived in the time-domain. The features, referred to as Multi-Frame Shifted Time Similarity (MFSTS), are obtained by computing constrained lag autocorrelations on overlapping speech frames to form a 2D map. A temporal convolutional neural network (TCN) [18] is then used to classify keywords on MFSTSs.

However, hand-crafted features such as MFSTSs and MFCCs may not be optimal for KWS. Therefore recent works have proposed to directly feed the DNNs with raw audio waveforms. In [19], a CNN for speaker recognition is proposed that encourages to learn parametrized sinc functions as kernels in the first layer. This layer is referred to as SincConv layer. During training, a low and high cutoff frequency per kernel is determined. Therefore, a custom filter bank is derived by training the SincConv layer that is specifically tailored to the desired application. SincConvs have also been recently applied to KWS tasks [20].

Quantization-aware training uses the straight-through estimator (STE) [21, 22] to approximate the gradient of piecewise constant quantizers by the non-zero gradient of some other